

COM3110/4115: Text Processing

Information Retrieval: Advanced Topics in IR

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- Definition of the information retrieval problem
- Approaches to document indexing
 - ◊ manual approaches
 - ◊ automatic approaches
- Automated retrieval models
 - ◊ boolean model
 - ◊ ranked retrieval methods (e.g. vector space model)
- Term manipulation:
 - ◊ stemming, stopwords, term weighting
- Evaluation
- Advanced Topics in IR

Advanced Topics in IR: Lecture Overview

- Using Metadata and Document Structure
- Alternative Approaches to Results Presentation
- Improving User Queries
 - ◊ Relevance Feedback
 - ◊ Query Expansion
- Web Search
 - ◊ PageRank
 - ◊ Query Log Analysis
- Neural Information Retrieval

What else do IR systems use

- Text is not only stored as unstructured information...

Jaguar

Mammal. The largest cat of the Americas, the jaguar has become a symbol of power, strength, and beauty. Once worshipped as a god, the jaguar's name comes from a native American word meaning "the killer that takes its prey in a single bound." Jaguars' skill in hunting is partly due to their extremely strong jaws. Jaguars make a variety of sounds, including roaring, mewling, and grunting. They are excellent swimmers, and they enjoy being in the water. While they do most of their stalking on the ground, they are also skilled at climbing and fishing. Like black leopards, black jaguars are called panthers.



Google Translate

Select Language

+ Google Gadgets powered by Google

Scientific Name	Lifespan
Panthera onca	12 years, 20 in captivity
Diet	

What else do IR systems use

- ... there is also metadata, e.g. in HTML format:

```
120 <div id="content">
121 <div class="h1">Jaguar</div>
122 <table width="508" border="0" cellpadding="8" cellspacing="0" class="outlinegray">
123   <tr bgcolor="#F3F3FC">
124     <td colspan="2" bgcolor="#FFFFFF"><table width="180" border="0" align="right" cellpadding="0" cellspacing="0">
125       <tr>
126         <td></td>
127       </tr>
128       <tr>
129         <td align="right"><script src="http://www.gmodules.com/ig/ifr?url=http://www.google.com/ig/modules/translateypage.xml&up_source_language=en&w=160&h=60&title=J
border=1output=js"></script></td>
130       </tr>
131     </table>
132     Mammal. The largest cat of the Americas, the jaguar has become a symbol of power, strength, and beauty. Once worshipped as a god, the jaguar's name comes from a native
American word meaning "the killer that takes its prey in a single bound." Jaguars' skill in hunting is partly due to their extremely strong jaws. Jaguars make a variety of
sounds, including roaring, mewling, and grunting. They are excellent swimmers, and they enjoy being in the water. While they do most of their stalking on the ground, they are also
skilled at climbing and fishing. Like black leopards, black jaguars are called panthers.</td>
133   </tr>
134   <tr bgcolor="#F3F3FC">
135     <td width="283" class="ruleright ruletopgray"><span class="textblue">Scientific
136       Name</span></td>
137     <td width="281" class="textblue ruletopgray">Lifespan</td>
138   </tr>
139   <tr>
140     <td class="ruleright">Panthera onca</td>
141     <td>12 years, 20 in captivity</td>
142   </tr>
```

- What from the HTML can be useful?
- What about RDF (Resource Description Framework) / OWL (Web Ontology Language), etc.? – **Semantic Web**

Parametric and zone indexes

- Most documents not only a sequence of terms
- Documents have multiple parts (or **zones**), some with special semantics:
 - ◇ Author
 - ◇ Title
 - ◇ Date of publication
 - ◇ Language
 - ◇ Format
 - ◇ etc.
- These constitute the **metadata** about a document

Parametric and zone indexes

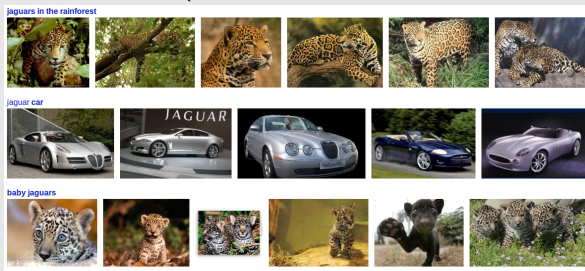
- We may wish to search by these metadata or these metadata + terms in the document
 - ◊ E.g., find docs authored by *William Shakespeare* in the year 1601, containing *Yorick*
Year = 1601
Author-Name = shakespeare
- Build inverted indexes on **zones** to allow structured queries
 - ◊ E.g. find docs with *merchant* in the title zone and where the text zone matches the query *gentle rain*
 - ◊ E.g. in Google:
https://developers.google.com/custom-search/docs/structured_search
- Different **weighting schemes** for terms:
 - ◊ Depending upon HTML field (e.g., **title**)
 - ◊ Depending upon display characteristics (e.g., **font**)

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Alternative Ways to Present Results: Visualisation & Clustering

- Return results by relevance only vs grouped somehow, e.g. by 'sense'
- Requires some notion of **similarity** between documents.
 - ◊ Could be the same metric used between q and D , or more advanced **clustering** methods
 - ◊ See: en.wikipedia.org/wiki/Document_clustering
- Clear to see with images (search 'jaguar' images on Google)



- Requires “topic” detection + labelling to describe grouped documents
- Not always provided by the search engine: extra layer on top of results

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Ways to Improve User Queries: Relevance Feedback

- In classical **relevance feedback** (Rocchio, 1971):
 - ◇ user frames an initial query
 - ◇ system returns an initial small set of documents
 - ◇ user specifies which of these documents are relevant/non-relevant
 - ◇ original query is reformulated according to distribution of terms in relevant and non-relevant documents that the user examined
 - ◇ new reformulated query is issued and further results returned – feedback loop may be iterated as many times as deemed useful

Ways to Improve User Queries: Relevance Feedback (cont)

- Formally:

$$\vec{q}_{i+1} = \vec{q}_i + \frac{\beta}{R} \sum_{j=1}^R \vec{r}_j - \frac{\gamma}{S} \sum_{k=1}^S \vec{s}_k$$

where

- ◇ R is the number of relevant documents returned by original query
- ◇ S is the number of non-relevant documents returned by original query
- ◇ $\vec{r}$ and $\vec{s}$ represent documents in the relevant and non-relevant sets, respectively
- ◇ β and γ , with $0 \leq \beta, \gamma \leq 1$ and $\beta + \gamma = 1$, are parameters controlling how much to weight the relevant versus non-relevant returned documents when reformulating the query

Ways to Improve User Queries: Relevance Feedback (cont)

- In **pseudo-relevance feedback** selection of relevant documents from initial list is automated
 - ◊ e.g. pick top n original ranked retrieval results and **assume** they are relevant
- Can also use, e.g., words in pages clicked by you or other users when searching for the same term(s) on previous occasions

Ways to Improve User Queries: Query expansion

- For a given query systems can suggest additional query words or phrases, e.g.:
 - ◊ synonyms
 - ◊ semantically related terms
- This is called **query expansion**
- Query expansion can be carried out
 - ◊ Using a manually created **thesaurus** – a resource containing word/phrases which are synonyms of each other
 - ◊ Using a thesaurus generated automatically for a specific collection by term clustering

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Web Search Ranking

- **Web docs** contain info beyond their mere *“textual content”*
 - ◇ *state-of-the-art web search* engines, like Google, exploit this
 - ◇ achieve *much more effective* retrieval than could without it
- **HTML** contains clues that some terms are *more important*
 - e.g. terms in regions marked as **title** or **headings**
 - e.g. terms *emphasised by formatting*: bold / bigger / colour
 - ◇ can use clever term weighting schemes, that add weight to such terms
- **Link text** — commonly provide *description of target* doc
 - ◇ often a better description than doc provides of *itself*
 - e.g. “Hey, here’s a great intro to calculus for beginners – check it out!”
 - ◇ Google treats **link text** as *part of* target doc
- **Link structure of web** *more broadly*
 - ◇ if page **A** *links to* page **B**, implies author of **A** thinks **B** worth looking at
 - ◇ can be used as a measure of *authority / quality*

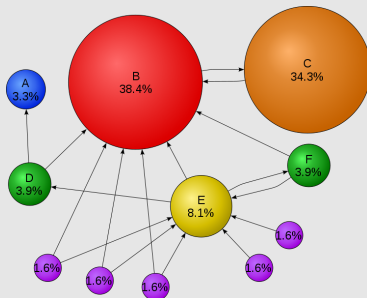
Exploiting Link Structure: the PageRank Algorithm

- Key method to exploit link structure of web: **PageRank algorithm**
 - ◇ named after its inventor: **Larry Page** (co-founder of Google)
 - ◇ assigns a score to each page on web – its *PageRank score*
 - can be seen to represent the page's *authority* (or *quality*)
- **PageRank algorithm** — key idea:
 - ◇ link from page A to page B confers **authority** on B
 - ◇ *how much* authority is conferred depends on:
 - the authority (PageRank score) of A, and its number of *out-going links*
i.e. A's authority is *shared out* amongst its out-going links
 - ◇ note that this measure is *recursively defined*
i.e. score of any page depends on score of every other page

- PageRank scores have an alternative interpretation:
 - ◇ probability that a *random surfer* will visit that page
 - i.e. one who starts at a random page, clicks randomly-chosen links forward, then (getting bored) jumps to a new random page, and so on ...
- Original algorithm:
 - ◇ Page A has *incoming links* $T_1 \dots T_n$
 - ◇ Page A has $C(A)$ links to other pages (*outgoing links*)
 - ◇ $0 \leq d \leq 1$ is a *damping factor*: probability that the person will stop clicking, usually $d = 0.85$
 - ◇ PageRank value for A : $PR(A) = (1 - d) + d * \left(\frac{PR(T_1)}{C(T_1)} + \dots + \frac{PR(T_n)}{C(T_n)} \right)$

Exploiting Link Structure: the PageRank algorithm (ctd)

- Graphical intuition:



- During retrieval, rank score of doc d is a *weighted combination* of:
 - its PageRank score: a measure of its authority
 - its IR-Score: how well d matches the query q , based on
 - Vector Space model, TF.IDF, *up-weighting* of important terms, etc.

Query Log Analysis

- Search engine **query logs** record significant amounts of information about each user session, including
 - ◇ client IP address
 - ◇ date, time and duration of session
 - ◇ query terms entered by user
 - ◇ results page returned
 - ◇ page(s) viewed by the user (“click thru”)
- This information can be analysed and used to improve search engine performance in a number of ways
 - ◇ data storage can be optimised to rapidly serve responses to common queries (e.g. via caching)
 - ◇ “click thru” data can be used to improve search results
 - as **relevance feedback** information
 - to train **learn-to-rank** algorithms

- Also used to build profiles of “users” and generate advertising revenue
- See:
 - ◇ B. J. Jansen (2006). Search log analysis: What it is, what’s been done, how to do it. *Library & Information Science Research*, 28, 407-432.
 - ◇ S.Zuboff (2019). *The Age of Surveillance Capitalism*. Profile Books.

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Neural IR: Word Embeddings

- In 2013 Tomas Mikolov and colleagues at Google published two papers defining an approach that has come to be called **word2vec** (see: <https://en.wikipedia.org/wiki/Word2vec>)
- Propose techniques to pre-train fixed length vectors for every word from very large corpora
 - ◇ Two approaches:
 - **CBOW** (continuous bag of words): train shallow feed forward neural net to predict missing target word w in middle of window of $2k + 1$ words
 - **Skip-gram**: train shallow feed forward neural net to predict words before and after target word w
 - ◇ In each case weights of pre-final layer of NN are extracted and become vector representation of word w , also called w 's **embedding**
 - ◇ Results show technique captures important aspects of word meaning.
 - Vectors of semantically similar words close to each other: **strong** $\sim$ **powerful**
 - Analogous semantic relationships are captured and can be demonstrated using vector arithmetic, e.g. **Paris - France + Italy = Rome** and **scientist - Einstein + Picasso = painter**

Neural IR: Word Embeddings (cont)

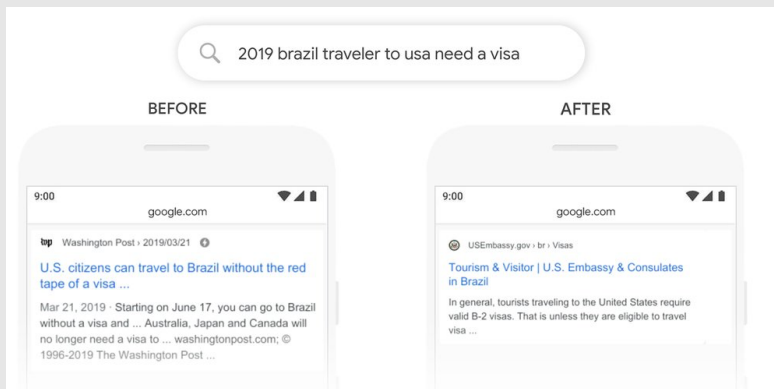
- Can these vector representations of words be used to improve search?
 - ◊ How do we represent a document or query, given vector representations of words?
- Various approaches tried
 - ◊ Represent word sequence (doc, query) as sum/average of word vectors
 - ◊ Represent word sequence as “bag of embedded words”
 - ◊ **doc2vec**: trains additional document vector in conjunction with word vectors in manner similar to word2vec (Le and Mikolov (2014). “Distributed Representations of Sentences and Documents”, ICML).
- First two approaches ignore word order, as does traditional vector space model
- doc2vec demonstrated superiority on several tasks but over-taken by further developments ...

- In 2018, Jacob Devlin and colleagues at Google published “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding”, aka simply **BERT**
(see [https://en.wikipedia.org/wiki/BERT_\(language_model\)](https://en.wikipedia.org/wiki/BERT_(language_model)))
 - ◇ word2vec offers representation of word meaning that is much richer than binary, tf or tfidf term weighting
 - ◇ BUT representation is context-free – all occurrences of word have same representation (cf. Tory party vs I like to party)
 - ◇ BERT extends idea of **word** embedding to let us create **sentence** embeddings, which allow us to capture use of words in context
- BERT was immediately shown to produce improvements beyond the state-of-the-art in a range of NLP tasks (not including IR)

- How can BERT be used for document retrieval?
 - ◇ Initial work applied BERT language models to re-ranking query results, treating the problem as a text classification problem
(B. Mitra and N. Craswell (2019). An introduction to neural information retrieval. Foundations and Trends in Information Retrieval, 13(1):1–126.)
 - ◇ Subsequent efforts looked at using BERT to aggregate sentence similarity scores between sentences in a candidate retrieved document and the query
(Yang et al. (2019) “Simple Applications of BERT for Ad Hoc Document Retrieval”)

Neural IR: Contextualised Word Embeddings (cont)

- In October 2019, Google announced they had started using BERT in their core search engine in the US
(www.blog.google/products/search/search-language-understanding-bert/)
- Claimed improves search results for 1 in 10 queries



Neural IR: Contextualised Word Embeddings (cont)

- How is Google doing it?
 - ◊ Not clear ...
- Many substantive engineering challenges
 - ◊ BERT output for single sentence contains contextualised embeddings for each word with, in its core model, 768 feature values per word of the input sentence.
 - ◊ Contextualised embeddings limited to 512 word input sequences (“sentences”)
 - How to represent documents that are multiple times this length?
 - ◊ How to index these complex document representations to allow relevant documents to be found to then compute similarity with queries?
 - Initial attempts show 1000-fold increase in index size ¹
 - ◊ Work going to address this, but remains an open question ...
- More on word2vec and BERT later in this module and in the spring semester NLP course

¹ opensourceconnections.com/blog/2019/11/05/understanding-bert-and-search-relevance/

Summary

- Effectiveness of search can be improved by taking into account **metadata** and **document structure** and allowing users to search within specified zones in documents.
- Results presentation may be improved by clustering documents that match a query into different groups and by, e.g., using images to differentiate the groups (e.g. “jaguar” the animal vs jaguar the car”)
- User queries may be automatically enhanced using techniques like **relevance feedback** and **query expansion**
- For hyper-linked collections like the Web, the link structure can be exploited by algorithms like **PageRank**
- User search behaviour can be exploited through **query log analysis**
- **Neural IR** is a new frontier in IR – expect exciting developments in this space